

HARSH GUPTA

VIT Bhopal University, Sehore | +91-7877454334 | hkgupta160420@gmail.com | LinkedIn | GitHub

EDUCATION

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science (Artificial Intelligence & Machine Learning)

Sep 2023 – Present

CGPA: **8.99/10.0** (Top 5%)

Senior School Certificate Examination, CBSE

2022 85%

Secondary School Examination, CBSE

2020 80%

EXPERIENCE

AI/ML Intern | EnviroWealth

Dec 2025 – Feb 2026

- Engineered an AI-powered BRSR ESG automation pipeline using Python, FastAPI, Gemini LLM, and Pandas. Extracted 400+ compliance fields from PDF/Excel via async REST APIs, cutting manual reporting effort by **~70%** and reducing turnaround time from days to hours.
- Designed a 3-layer NLP transformation architecture with chunked Gemini API prompting across 8 document sections. Built a custom flat-to-nested JSON parser for structured, scalable compliance output, ensuring consistent data integrity across all reporting categories.
- Implemented MongoDB data storage and Dockerized full-stack deployment on Vercel, delivering a reproducible, production-grade ESG reporting automation system with end-to-end traceability and zero manual intervention in the data pipeline.

PROJECTS

Advanced Brain Cancer Detection Using CNN Architectures

[GitHub]

- Implemented an ensemble of CNN models (VGG16, MobileNet, ResNet50) using Keras and TensorFlow, achieving **99.8% classification accuracy** on the BraTS2020 dataset through weighted voting and feature fusion across all three architectures, significantly outperforming any single-model baseline.
- Preprocessed 1,000+ MRI images using OpenCV, applying normalization, augmentation, and skull-stripping techniques to improve model generalizability; deployed a clinical-grade interface with Flask containerization for real-time diagnostic use.
- Authored a 7-page peer-reviewed research paper documenting end-to-end methodology, ablation studies, and benchmark comparisons under faculty supervision, with findings submitted to a refereed academic conference.

VaidyaSetu — AI Healthcare Prototype

- Built a full-stack AI healthcare prototype integrating telemedicine via WebRTC, OCR-LLM pipelines using Tesseract and EasyOCR, and AI-driven diagnostics, improving patient accessibility by **+35%** across a 20-user pilot cohort by eliminating geographic barriers to specialist consultation.
- Engineered an OCR-LLM pipeline integrated with the Google Calendar API to automate medical report extraction, field parsing, and appointment scheduling, cutting manual administrative effort by **~60%** and improving scheduling accuracy for follow-up visits.
- Trained a CNN classification model on 10,000+ labeled medical images achieving **96% diagnostic accuracy**; deployed an NLP-powered chatbot for patient-facing report summarization, improving health comprehension scores by **+40%** in post-pilot surveys.

LLaMA-3 QLoRA Fine-Tuning for Heart Disease Recommendations

[Colab]

- Fine-tuned LLaMA-3 8B using QLoRA + LoRA adapter methods with Hugging Face Transformers, reducing GPU memory consumption by **50%** while preserving full model accuracy, enabling training on consumer-grade hardware without performance trade-offs.
- Automated preprocessing, tokenization, and publishing of 2,000+ structured JSON records to Hugging Face Datasets Hub, cutting data preparation effort by **70%** and enabling reproducible dataset versioning for downstream model evaluation.
- Optimized the end-to-end training pipeline using TRL, Accelerate, and bitsandbytes libraries with gradient checkpointing, AdamW-8bit optimizer, and mixed-precision training, improving overall training efficiency by **30%** while reducing peak VRAM usage significantly.

ACHIEVEMENTS

- Winner** — VIT Mother's Day Challenge; designed an innovative solution driving engagement among 100+ participants.
- 3rd Place** — Hack4Health Hackathon at VIT Bhopal.

TECHNICAL SKILLS

AI / ML Scikit-learn, Pandas, NumPy, TensorFlow, Keras, Hugging Face, RAG, Generative AI, LLMs, CNNs, EfficientNet, XGBoost, LightGBM

LLM / Agents LangChain, LangGraph, LangSmith, llama.cpp, LlamaIndex, QLoRA, LoRA, Unsloth, Transformers, PEFT, TRL

CV / OCR OpenCV, Tesseract, EasyOCR, MobileNet, ResNet, VGG16

Programming Python (DSA), Java, C++, HTML, CSS, MySQL

Cloud/DevOps Docker, Git, GitHub, FastAPI, Flask, Streamlit, MongoDB, Render, Vercel, Postman

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate**
- Generative AI using IBM Watsonx** — IBM Developer Skill Network
- MATLAB and Simulink Technical Computing Certification** — MathWorks
- GeeksforGeeks 160-Day DSA Mastery Certificate** — GeeksforGeeks

Jul 2025

Jun 2025